SERVICENOW SYSTEM ADMINISTRATOR · NAAN MUDHALVAN (SMARTBRIDGE)

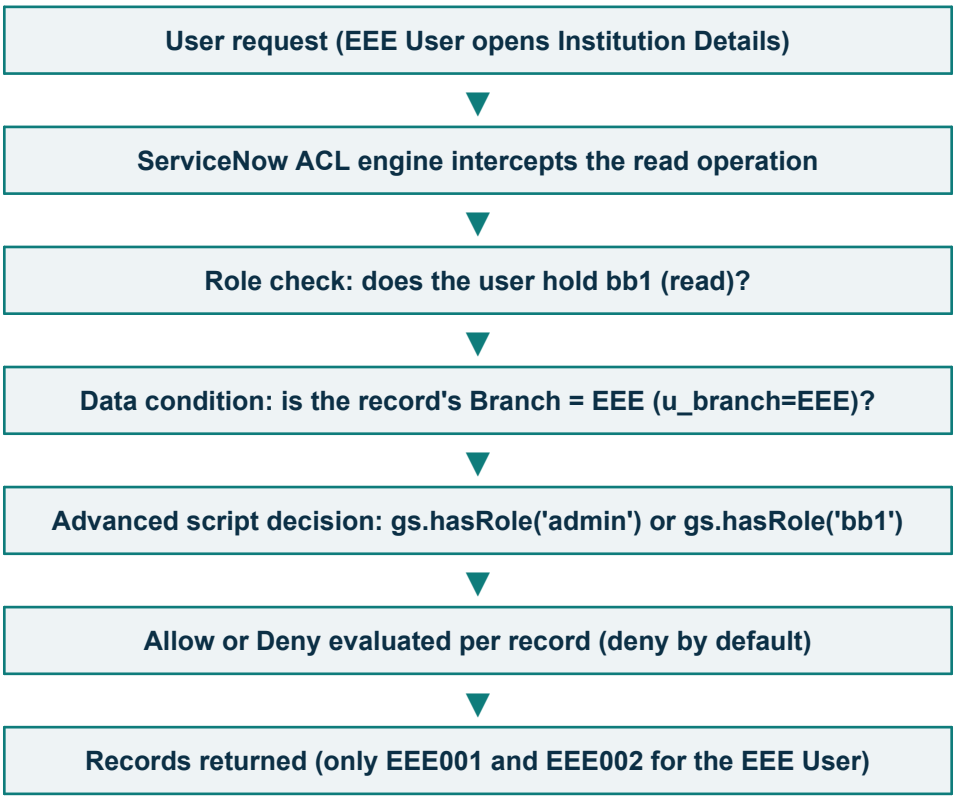
Data Flow Diagram - Field-Value-Driven ACL Evaluation

Phase 2 · Requirement Analysis

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This document describes how a data request flows through the ServiceNow Access Control (ACL) engine for the u_institution_details table, and which platform data stores participate in the decision. Access is decided by both the user's role and the Branch field value on each record, so the same request returns different results for the EEE User, an administrator, and a user with no roles.

Request Flow



The engine evaluates the read ACL for every candidate record. A record passes only when the role requirement, the Branch=EEE condition, and the script all resolve to true; an administrator passes on the script's admin branch and therefore sees all records, while a user lacking bb1 is denied and shown the security-constraints page. The create, write, and delete operations follow the same interception pattern against the bb2, bb3, and bb4 roles.

Data Stores Involved

Data Store (Table)	Role in the Flow
u_institution_details	Custom table holding the student records (roll number, name, faculty, Branch, email, phone, description) that are filtered per request
sys_security_acl	Stores the four ACL definitions (read, create, write, delete), including the READ rule's Advanced flag, the Branch=EEE condition, and the GlideScript

Data Store (Table)	Role in the Flow
sys_security_acl_role	Links each ACL to its required role (bb1 read, bb2 create, bb3 write, bb4 delete)
sys_user_has_role	Maps users to roles; confirms the EEE User holds bb1-bb4 during evaluation and impersonation testing